

JalDrishti: Geo-Visualization for Watershed Intelligence

Smart India Hackathon 2026 — Project Report

Team: Genesis X | **Problem Statement ID:** SIH26015 | **Theme:** Agriculture, FoodTech & Rural Development | **PS Category:** Software

1. Problem Statement

Title: Application of Geospatial Techniques for Visualization and Analysis to Interpret Geo-Coded Images and Enhance Watershed Development Outcomes.

Watershed development programs currently suffer from four core gaps:

- **Fragmented monitoring** — field data and satellite data exist in silos with no integrated view.
- **Manual and slow processes** — geo-tagged photos are collected but used only for documentation, not analysis.
- **No standard framework** — planners lack a consistent way to convert geo-coded images into actionable insight.
- Interventions are tracked passively rather than evaluated against measurable, evidence-based outcomes.

2. Proposed Solution — JalDrishti

JalDrishti ("Water Vision") is positioned as the first framework to fuse geo-tagged field photographs with SRISHTI-DRISHTI satellite data (30m resolution) into a single, interpretable dashboard, purpose-built to strengthen India's existing SRISHTI-DRISHTI platform.

Core capabilities:

- **Integrated Visualization Engine** — merges field geo-photos with satellite thematic layers on one map.
- **Automated Thematic Mapping** — auto-generates land use, drainage, vegetation, and water-body maps from raw imagery.
- **Change Detection Module** — before/after NDVI comparison to quantify real impact of watershed interventions.
- **Geo-Tag Intelligence** — automatically extracts GPS coordinates and timestamps from field photos (no manual entry).

Unique Value Proposition:

Aspect	Description
First of its kind	Combines field photos + satellite layers in one dashboard
Lightweight & scalable	Runs on an open-source GIS stack, replicable to any Indian watershed
Evidence-based	Converts scattered photos into quantifiable, comparable spatial data
Temporal	Enables before-and-after comparison of change
Built for SRISHTI-DRISHTI	Designed to extend the existing government platform, not replace it

The dashboard is organized around six perspectives — Watershed Landscape, Geo-tagged Field Evidence, Satellite Perspective, Temporal Change, Spatial Context, and an overview hub — moving the workflow **from**

scattered photos and satellite data → integrated automated analysis → smart spatial insight → stronger watersheds → sustainable, better decisions.

3. Technical Approach

Data Inputs

- Satellite imagery — Resourcesat-2 LISS-III & 1m multispectral (B2, B3, B4, B5)
- Terrain data — CartoDEM (~32m resolution)
- Thematic data — LULC, drainage, water bodies
- Field evidence — geo-tagged photographs (location, activity, date, status)
- Supporting data — IMD rainfall data, soil data

Processing Pipeline

- Data acquisition from multiple sources
- Preprocessing (format conversion, noise removal, band selection)
- Georeferencing & CRS alignment (WGS84/UTM)
- Watershed/AOI clipping
- Parallel feature analysis — NDVI (vegetation), NDWI (surface water), Slope (terrain/runoff), Drainage (flow network), LULC (land cover), Temporal Change
- Spatial correlation of field observations with satellite, terrain, and hydrological context
- Watershed priority/suitability modeling via multi-criteria spatial analysis
- GIS visualization — integration into an interactive dashboard
- Decision support — evidence-based insights for planning, monitoring, and evaluation

Technology Stack

Layer	Tools
Geospatial processing	QGIS, GDAL, Rasterio, GeoPandas, Shapely
Analytics / modeling	Python, NumPy, Pandas, Scikit-learn
Backend	FastAPI (Python)
Database	PostgreSQL, PostGIS
Frontend / visualization	React, Leaflet, MapLibre
Deployment	Docker

Core Analytical Outputs: Vegetation Condition, Land-Use Pattern, Water Availability, Temporal Change, Terrain/Runoff, and Intervention Context.

4. Feasibility and Viability

Feasibility Pillars

- **Technical** — built entirely on a proven open-source geospatial stack (Rasterio/GeoPandas); no proprietary licensing cost.
- **Data access** — SRISHTI-DRISHTI already provides free 30m resolution data via ISRO/Bhuvan.
- **Operational** — runs on standard field-office hardware; no specialized GPU or server needed.

- **Economic** — zero recurring software cost, scalable to any number of watersheds.

Viability Pillars

- **Policy fit** — directly supports the Watershed Development Component (WDC-PMKSY) and Jal Shakti Abhiyan.
- **Government utility** — ready for adoption by DRDA/District Watershed Committees for monitoring.
- **Scalability** — watershed-agnostic; replicable across regions with minimal reconfiguration.
- **Revenue / sustainability** — deployable as a SaaS-style GIS module for state rural development departments.

Risks & Mitigations

Risk	Mitigation
Cloud cover / seasonal variation affecting satellite image quality	Multi-temporal compositing and cloud-masking algorithms
Field staff unfamiliar with GIS tools may resist adoption	Simple one-click interface requiring no GIS training

Build path: Open Data → Open GIS → Modular System → Scalable Watershed Monitoring, developed across four stages — open data access, open-source GIS implementation, modular system design, and a prototype deployed on a selected watershed.

5. Impact and Benefits

Dimension	Impact
Economic	Better-targeted watershed investments reduce wasted expenditure on ineffective interventions; supports data-backed fund allocation under PMKSY.
Environmental	Early detection of land degradation and drainage issues enables faster corrective action, improving groundwater recharge and vegetation cover.
Social & Governance	Transparent, evidence-based reporting builds trust between implementing agencies and beneficiary communities; supports Digital India's e-governance push.

Comparison — Today vs. JalDrishti

Feature	Today	JalDrishti
Field photo usage	Documentation only	Integrated spatial analysis
Satellite + field data fusion	Manual / None	Automated
Change detection	Rare / manual	Automated, NDVI-based
Cost	High (manual surveys)	Low (open-source stack)
Scalability	Limited	High, replicable

6. Research and References

The **Watershed Analytics Framework** draws on six foundational domains: geospatial libraries (efficient geospatial data manipulation), Python libraries (data analysis/processing), vegetation indices (vegetation health and water content), terrain analysis (surface characteristics), land use analysis (classification/mapping of land cover types), and temporal analysis (tracking change over time) — together producing a comprehensive geospatial understanding of the watershed.

Key Resources

Resource	Source	Link
Geo-tagged Field Photos	Bhuvan SRISHTI-DRISHTI	bhuvanapp1.nrsc.gov.in/iwmp
Satellite Imagery	Bhoonidhi — ISRO EO Data Hub	—

Summary

JalDrishti proposes to close the loop between ground-truth field evidence and satellite-based monitoring for India's watershed development programs. By automatically extracting geo-tags from field photographs and fusing them with SRISHTI-DRISHTI's 30m satellite layers (NDVI, NDWI, slope, drainage, LULC), the platform converts what is currently a documentation exercise into a quantifiable, temporal, decision-support system — built entirely on an open-source, low-cost stack designed for direct integration with existing PMKSY / Jal Shakti Abhiyan monitoring workflows.